

Lengthy Derivations in Sect. 6.3.1

Basic Integrals (1)

2.26 Forms containing $\sqrt{a + bx + cx^2}$ and integral powers of x

Notation: $R = a + bx + cx^2$, $\Delta = 4ac - b^2$

$$\begin{aligned}
 2.261^{11} \quad \text{For } n = -1 \quad & \int \frac{dx}{\sqrt{R}} = \frac{1}{\sqrt{c}} \ln \left(\frac{2\sqrt{cR} + 2cx + b}{\sqrt{\Delta}} \right) \quad [c > 0] \\
 & = \frac{1}{\sqrt{c}} \operatorname{arcsinh} \left(\frac{2cx + b}{\sqrt{\Delta}} \right) \quad [c > 0, \quad \Delta > 0] \\
 & = \frac{1}{\sqrt{c}} \ln(2cx + b) \quad [c > 0, \quad \Delta = 0] \\
 & = \frac{-1}{\sqrt{-c}} \arcsin \left(\frac{2cx + b}{\sqrt{-\Delta}} \right) \quad [c < 0, \quad \Delta < 0]
 \end{aligned}$$

$$\begin{aligned}
 5. \quad & \int \frac{dx}{\sqrt{R^3}} = \frac{2(2cx + b)}{\Delta \sqrt{R}} \\
 6. \quad & \int \frac{x dx}{\sqrt{R^3}} = -\frac{2(2a + bx)}{\Delta \sqrt{R}} \\
 7. \quad & \int \frac{x^2 dx}{\sqrt{R^3}} = -\frac{(\Delta - b^2)x - 2ab}{c\Delta \sqrt{R}} + \frac{1}{c} \int \frac{dx}{\sqrt{R}} \quad (\text{see 2.261}) \\
 8. \quad & \int \frac{x^3 dx}{\sqrt{R^3}} = \frac{c\Delta x^2 + b(10ac - 3b^2)x + a(8ac - 3b^2)}{c^2 \Delta \sqrt{R}} - \frac{3b}{2c^2} \int \frac{dx}{\sqrt{R}} \\
 & \quad \quad \quad (\text{see 2.261})
 \end{aligned}$$

$$Q = 1 + x^2 - 2x \cos \psi$$

$$a = 1, \quad b = -2 \cos \psi, \quad c = 1, \quad \Delta = 4(1 - \cos^2 \psi) = 4 \sin^2 \psi$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{4(1 - \cos^2 \psi - \cos^2 \psi)x + 4 \cos \psi}{4(1 - \cos^2 \psi)Q^{\frac{1}{2}}} + \ln \left(\frac{2Q^{\frac{1}{2}} + 2x - 2 \cos \psi}{2 \sin \psi} \right) + C$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} + \ln \left(\frac{Q^{\frac{1}{2}} + x - \cos \psi}{\sin \psi} \right) + C$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} + \ln \left(\frac{Q^{\frac{1}{2}} + x - \cos \psi}{\sin \psi} \right) + C$$

Basic Integrals (2)

$$u = \sqrt{a + cx^2}$$

$$I_1 = \frac{1}{\sqrt{c}} \ln(x\sqrt{c} + u)$$

$$\int \frac{dx}{u} = I_1$$

$$\int \frac{dx}{u^3} = \frac{1}{a} \frac{x}{u}$$

$$\int \frac{dx}{u^{2n+1}} = \frac{1}{a^n} \sum_{k=0}^{n-1} \frac{(-1)^k}{2k+1} \binom{n-1}{k} \frac{c^k x^{2k+1}}{u^{2k+1}}$$

$$\int \frac{x dx}{u^{2n+1}} = -\frac{1}{(2n-1)cu^{2n-1}}$$

$$a = 2 - 2 \cos \psi, \quad c = 1$$

$$\int \frac{d\delta x}{(2 - 2 \cos \psi + \delta x^2)^{1/2}} = \ln[\delta x + (2 - 2 \cos \psi + \delta x^2)^{1/2}] + C$$

$$\int \frac{\delta x d\delta x}{(2 - 2 \cos \psi + \delta x^2)^{1/2}} = (2 - 2 \cos \psi + \delta x^2)^{1/2} + C$$

$$\int \frac{\delta x d\delta x}{(2 - 2 \cos \psi + \delta x^2)^{3/2}} = -\frac{1}{(2 - 2 \cos \psi + \delta x^2)^{1/2}} + C$$

$$\int \frac{\delta x d\delta x}{(2 - 2 \cos \psi + \delta x^2)^{3/2}} = -\frac{1}{(2 - 2 \cos \psi + \delta x^2)^{1/2}} + C$$

$$\int \frac{d\delta x}{(2 - 2 \cos \psi + \delta x^2)^{3/2}} = \frac{\delta x}{(2 - 2 \cos \psi)(2 - 2 \cos \psi + \delta x^2)^{1/2}} + C$$

$$\int \frac{\delta x d\delta x}{(2 - 2 \cos \psi + \delta x^2)^{5/2}} = -\frac{1}{3(2 - 2 \cos \psi + \delta x^2)^{3/2}} + C$$

Eq. (6.61)

(6.61) is straightforward with integral formula

$$\begin{aligned} \int \frac{x^2 dx}{(1+x^2-2x \cos \psi)^{1/2}} \\ = \left(\frac{x}{2} + \frac{3 \cos \psi}{2} \right) (1+x^2-2x \cos \psi)^{1/2} \\ + \left(\frac{3 \cos^2 \psi}{2} - \frac{1}{2} \right) \ln [(1+x^2-2x \cos \psi)^{1/2} + x - \cos \psi] + C \end{aligned}$$

This formula is from the following formula copied from page 16 of this document:

$$\begin{aligned} \int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x}{2} + \frac{3 \cos \Theta}{2} \right) (1+x^2-2x \cos \Theta)^{\frac{1}{2}} \\ + \left(\frac{3 \cos^2 \Theta}{2} - \frac{1}{2} \right) \ln \left[(1+x^2-2x \cos \Theta)^{\frac{1}{2}} + x - \cos \Theta \right] + C \end{aligned}$$

Eq. (6.65)—(6.67)

$$\begin{aligned} f(x) = (x + 3 \cos \psi)(1+x^2-2x \cos \psi)^{1/2} \\ + (3 \cos^2 \psi - 1) \ln [(1+x^2-2x \cos \psi)^{1/2} + x - \cos \psi] \end{aligned}$$

$$\begin{aligned} f'(x) &= (1+x^2-2x \cos \psi)^{1/2} + \frac{(x+3 \cos \psi)(x-\cos \psi)}{(1+x^2-2x \cos \psi)^{1/2}} \\ &+ \frac{(3 \cos^2 \psi - 1) \left[\frac{x-\cos \psi}{(1+x^2-2x \cos \psi)^{1/2}} + 1 \right]}{(1+x^2-2x \cos \psi)^{1/2} + x - \cos \psi} \\ &= (1+x^2-2x \cos \psi)^{1/2} + \frac{x^2 - x \cos \psi + 3x \cos \psi - 3 \cos^2 \psi}{(1+x^2-2x \cos \psi)^{1/2}} \\ &+ \frac{3 \cos^2 \psi - 1}{(1+x^2-2x \cos \psi)^{1/2}} \\ &= (1+x^2-2x \cos \psi)^{1/2} + \frac{x^2 + 2x \cos \psi - 3 \cos^2 \psi + (3 \cos^2 \psi - 1)}{(1+x^2-2x \cos \psi)^{1/2}} \\ &= (1+x^2-2x \cos \psi)^{1/2} + \frac{x^2 + 2x \cos \psi - 1}{(1+x^2-2x \cos \psi)^{1/2}} \\ &= (1+x^2-2x \cos \psi)^{1/2} + \frac{2x^2 + 2x \cos \psi - 1 - x^2}{(1+x^2-2x \cos \psi)^{1/2}} \\ &= (1+x^2-2x \cos \psi)^{1/2} + \frac{2x \cos \psi - 1 - x^2}{(1+x^2-2x \cos \psi)^{1/2}} + \frac{2x^2}{(1+x^2-2x \cos \psi)^{1/2}} \\ &= \frac{2x^2}{(1+x^2-2x \cos \psi)^{1/2}} \end{aligned}$$

$$\begin{aligned}
f''(x) &= \frac{4x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{2x^2(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&= \frac{4x(1+x^2-2x\cos\psi) - 2x^2(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&= \frac{4x + 4x^3 - 8x^2\cos\psi - 2x^3 + 2x^2\cos\psi}{(1+x^2-2x\cos\psi)^{3/2}} \\
&= \frac{4x + 2x^3 - 6x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}
\end{aligned}$$

$$\begin{aligned}
f'''(x) &= \frac{4+6x^2-12x\cos\psi}{(1+x^2-2x\cos\psi)^{3/2}} - \frac{3(4x+2x^3-6x^2\cos\psi)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4+6x^2-12x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3(3x+3x^3-6x^2\cos\psi+x-x^3)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4+6x^2-12x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3(3x+3x^3-6x^2\cos\psi)(2x-2\cos\psi) + 3(x-x^3)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4+6x^2-12x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3(3x+3x^3-6x^2\cos\psi)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{3(x-x^3)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4+6x^2-12x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{9x(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3(x-x^3)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4+6x^2-12x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{9x^2-9x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{(x-x^3)(2x-2\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4+6x^2-12x\cos\psi-9x^2+9x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3(x-x^3)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{4-3x^2-3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3(x-x^3)(2x-2\cos\psi)}{2(1+x^2-2x\cos\psi)^{\frac{5}{2}}}
\end{aligned}$$

$$\begin{aligned}
&= \frac{8 - 6x^2 - 6x \cos \psi}{2(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - \frac{3x(1 - x^2)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\
&= \frac{(3 + 3x^2 - 6x \cos \psi) + 5 - 9x^2}{2(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - \frac{3x(1 - x^2)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\
&= \frac{3 + 3x^2 - 6x \cos \psi}{2(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} + \frac{5 - 9x^2}{2(1 + x^2 - 2x \cos \psi)^{3/2}} - \frac{3x(1 - x^2)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{5/2}} \\
&= \frac{3}{2(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + \frac{5 - 9x^2}{2(1 + x^2 - 2x \cos \psi)^{3/2}} - \frac{3x(1 - x^2)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{5/2}}
\end{aligned}$$

Eq. (6.85)

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{4(1 - \cos^2 \psi - \cos^2 \psi)x + 4 \cos \psi}{4(1 - \cos^2 \psi)Q^{1/2}} + \ln \left(\frac{2Q^{1/2} + 2x - 2 \cos \psi}{2 \sin \psi} \right)$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi}{(1 - \cos^2 \psi)Q^{1/2}} + \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right)$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi}{(1 - \cos^2 \psi)Q^{1/2}} + \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right)$$

$$\begin{aligned}
\int \frac{x^3 dx}{Q^{3/2}} &= \frac{4(1 - \cos^2 \psi)x^2 - 2 \cos \psi (10 - 12 \cos^2 \psi)x + 8 - 12 \cos^2 \psi}{4(1 - \cos^2 \psi)Q^{1/2}} \\
&\quad + 3 \cos \psi \ln \left(\frac{2Q^{1/2} + 2x - 2 \cos \psi}{2 \sin \psi} \right)
\end{aligned}$$

$$\begin{aligned}
\int \frac{x^3 dx}{Q^{3/2}} &= \frac{(1 - \cos^2 \psi)x^2 - \cos \psi (5 - 6 \cos^2 \psi)x + 2 - 3 \cos^2 \psi}{(1 - \cos^2 \psi)Q^{1/2}} \\
&\quad + 3 \cos \psi \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right)
\end{aligned}$$

$$\begin{aligned}
& \int \left(\frac{x^2}{Q^{3/2}} - \cos \psi \frac{x^3}{Q^{3/2}} \right) dx \\
&= -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} + \ln \left(\frac{Q^{\frac{1}{2}} + x - \cos \psi}{\sin \psi} \right) \\
&- \cos \psi \frac{(1 - \cos^2 \psi)x^2 - \cos \psi (5 - 6 \cos^2 \psi)x + 2 - 3 \cos^2 \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&- 3 \cos^2 \psi \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right)
\end{aligned}$$

$$\begin{aligned}
& \int \left(\frac{x^2}{Q^{3/2}} - \cos \psi \frac{x^3}{Q^{3/2}} \right) dx \\
&= -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi + \cos \psi (1 - \cos^2 \psi)x^2 - \cos^2 \psi (5 - 6 \cos^2 \psi)x + 2 \cos \psi - 3 \cos^3 \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&+ (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= -\frac{(1 - 2 \cos^2 \psi)x + \cos \psi (1 - \cos^2 \psi)x^2 - \cos^2 \psi (5 - 6 \cos^2 \psi)x + 3 \cos \psi - 3 \cos^3 \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= -\frac{(1 - \cos^2 \psi)x - x \cos^2 \psi + \cos \psi (1 - \cos^2 \psi)x^2 - \cos^2 \psi (5 - 6 \cos^2 \psi)x + 3 \cos \psi - 3 \cos^3 \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&+ (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= -\frac{(1 - \cos^2 \psi)x + \cos \psi (1 - \cos^2 \psi)x^2 - x \cos^2 \psi - \cos^2 \psi (5 - 6 \cos^2 \psi)x + 3 \cos \psi - 3 \cos^3 \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&+ (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= -\frac{(1 - \cos^2 \psi)x + \cos \psi (1 - \cos^2 \psi)x^2 - \cos^2 \psi (6 - 6 \cos^2 \psi)x + 3 \cos \psi - 3 \cos^3 \psi}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&+ (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{\frac{1}{2}} + x - \cos \psi}{\sin \psi} \right)
\end{aligned}$$

$$\begin{aligned}
&= - \frac{(1 - \cos^2 \psi)x + \cos \psi (1 - \cos^2 \psi)x^2 - 6 \cos^2 \psi (1 - \cos^2 \psi)x + 3 \cos \psi (1 - \cos^2 \psi)}{(1 - \cos^2 \psi)Q^{\frac{1}{2}}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{\frac{1}{2}} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{x + x^2 \cos \psi - 6 x \cos^2 \psi + 3 \cos \psi}{Q^{\frac{1}{2}}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{x - 2x^2 \cos \psi + 3x^2 \cos \psi - 6 x \cos^2 \psi + 3 \cos \psi}{Q^{\frac{1}{2}}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{x - 2x^2 \cos \psi + 3 \cos \psi (x^2 - 2x \cos \psi + 1)}{Q^{\frac{1}{2}}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{x - 2x^2 \cos \psi + 3 \cos \psi Q}{Q^{\frac{1}{2}}} + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{x - 2x^2 \cos \psi}{Q^{\frac{1}{2}}} - 3 \cos \psi Q^{\frac{1}{2}} + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{-x^3 + x^3 + x - 2x^2 \cos \psi}{Q^{\frac{1}{2}}} - 3 \cos \psi Q^{\frac{1}{2}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{-x^3 + x(x^2 + 1 - 2x \cos \psi)}{Q^{\frac{1}{2}}} - 3 \cos \psi Q^{\frac{1}{2}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= - \frac{-x^3 + xQ}{Q^{\frac{1}{2}}} - 3 \cos \psi Q^{\frac{1}{2}} + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= \frac{x^3}{Q^{\frac{1}{2}}} - xQ^{\frac{1}{2}} - 3 \cos \psi Q^{\frac{1}{2}} + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{x^3}{Q^{\frac{1}{2}}} - (x + 3 \cos \psi) Q^{\frac{1}{2}} + (1 - 3 \cos^2 \psi) \ln \left(\frac{Q^{1/2} + x - \cos \psi}{\sin \psi} \right) \\
&= \frac{x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - (x + 3 \cos \psi)(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left[\frac{(1 + x^2 - 2x \cos \psi)^{1/2} + x - \cos \psi}{\sin \psi} \right]
\end{aligned}$$

Eq. (6.90)—(6.92)

$$\begin{aligned}
f(x) &= \frac{x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - (x + 3 \cos \psi)(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&\quad + (1 - 3 \cos^2 \psi) \ln \left[(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} + x - \cos \psi \right] \\
f'(x) &= \frac{3x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&\quad - \frac{(x + 3 \cos \psi)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + \frac{(1 - 3 \cos^2 \psi) \left[\frac{x - \cos \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + 1 \right]}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} + x - \cos \psi} \\
&= \frac{3x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&\quad - \frac{(x + 3 \cos \psi)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + \frac{(1 - 3 \cos^2 \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} \\
&= \frac{3x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + \frac{(1 - 3 \cos^2 \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{(x + 3 \cos \psi)(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} \\
&\quad - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&= \frac{3x^2 + 1 - 3 \cos^2 \psi - x^2 + x \cos \psi - 3x \cos \psi + 3 \cos^2 \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} \\
&\quad - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}
\end{aligned}$$

$$\begin{aligned}
f'(x) &= \frac{2x^2 + 1 - 2x \cos \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&= \frac{x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + \frac{x^2 + 1 - 2x \cos \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
&\quad - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} \\
&= \frac{x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} + (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
&\quad - (1 + x^2 - 2x \cos \psi)^{\frac{1}{2}} = \frac{x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}}
\end{aligned}$$

$$\begin{aligned}
f''(x) &= \frac{2x}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^2(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - \frac{3x^2(x - \cos \psi) + x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
&\quad + \frac{3x^3(x - \cos \psi)^2}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\
&= \frac{2x}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{4x^2(x - \cos \psi) + x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} + \frac{3x^3(x - \cos \psi)^2}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}}
\end{aligned}$$

$$\begin{aligned}
f'''(x) &= \frac{2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{2x(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - \frac{8x(x - \cos \psi) + 4x^2 + 3x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
&\quad + \frac{12x^2(x - \cos \psi)^2 + 3x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} + \frac{9x^2(x - \cos \psi)^2 + 6x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\
&\quad - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{10x(x - \cos \psi) + 7x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x - \cos \psi)^2 + 9x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}}
\end{aligned}$$

Eq. (6.90)—(6.92)

alternative using result (6.65)—(6.67)

$$f(x) = \frac{x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}}$$

$$f'(x) = \frac{3x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}}$$

$$\begin{aligned} f''(x) &= \frac{6x}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{3x^2(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - \frac{3x^2(x - \cos \psi) + x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\ &\quad + \frac{3x^3(x - \cos \psi)^2}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\ &= \frac{6x}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{6x^2(x - \cos \psi) + x^3}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} + \frac{3x^3(x - \cos \psi)^2}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \end{aligned}$$

$$\begin{aligned} f'''(x) &= \frac{6}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{6x(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} - \frac{12x(x - \cos \psi) + 6x^2 + 3x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\ &\quad + \frac{18x^2(x - \cos \psi)^2 + 3x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} + \frac{9x^2(x - \cos \psi)^2 + 6x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\ &\quad - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}} \\ &= \frac{6}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{18x(x - \cos \psi) + 9x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\ &\quad + \frac{27x^2(x - \cos \psi)^2 + 9x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}} \end{aligned}$$

XXXXXXXXXXXXXXXXXXXXXXXXXX

$$\begin{aligned} & \frac{3x^2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{2x^2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} \\ &= \frac{x^2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \end{aligned}$$

$$\begin{aligned} & \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{6x^2(x-\cos\psi)+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\ & \quad - \frac{4x+2x^3-6x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\ &= \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{6x^2(x-\cos\psi)+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{4x+2x^3-6x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\ & \quad + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\ &= \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{6x^3-6x^2\cos\psi+x^3+4x+2x^3-6x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\ & \quad + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\ &= \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{9x^3-12x^2\cos\psi+4x}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(1-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\ &= \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x+4x^3-8x^2\cos\psi+5x^3-4x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\ & \quad + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\ &= \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x(1+x^2-2x\cos\psi)+5x^3-4x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\ & \quad + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\ &= \frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{5x^3-4x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\ & \quad + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \end{aligned}$$

$$\begin{aligned}
&= \frac{2x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x^3-4x^2\cos\psi+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{2x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x^2(x-4\cos\psi)+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&\frac{6}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{18x(x-\cos\psi)+9x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{27x^2(x-\cos\psi)^2+9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&\quad - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&\quad - \left[\frac{3}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} + \frac{5-9x^2}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3x(1-x^2)(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \right] \\
&= \frac{6}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{3}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{18x(x-\cos\psi)+9x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad - \frac{5-9x^2}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{27x^2(x-\cos\psi)^2+9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&\quad + \frac{3x(1-x^2)(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{36x(x-\cos\psi)+5+9x^2}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2+[9x^3+3x(1-x^2)](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{36x^2-36x\cos\psi+5+9x^2}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2+[9x^3+3x-3x^3](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{5+5x^2-10\cos\psi+40x^2-26x\cos\psi}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2+[6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}
\end{aligned}$$

$$\begin{aligned}
&= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{5}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{40x^2-26x\cos\psi}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{40x^2-26x\cos\psi}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{20x^2-13x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi) + 7x^2 + 3x^2 - 3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi) + 7x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3x^2-3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x-\cos\psi)^2 + 9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&\quad + \frac{6x^2(x-\cos\psi)^2 + [3x-3x^3](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi) + 7x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x-\cos\psi)^2 + 9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&\quad - \frac{3x^2-3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{6x^2(x-\cos\psi)^2 + [3x-3x^3](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi) + 7x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x-\cos\psi)^2 + 9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3x^2 - 3x \cos \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} + \frac{[6x^3 - 6x^2 \cos \psi + 3x - 3x^3](x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\
& = \frac{2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{10x(x - \cos \psi) + 7x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
& + \frac{21x^2(x - \cos \psi)^2 + 9x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}} \\
& - \frac{3x^2 - 3x \cos \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} + \frac{[3x^3 - 6x^2 \cos \psi + 3x](x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} \\
& = \frac{2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{10x(x - \cos \psi) + 7x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
& + \frac{21x^2(x - \cos \psi)^2 + 9x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}} \\
& - \frac{3x^2 - 3x \cos \psi}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} + \frac{3x(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
& = \frac{2}{(1 + x^2 - 2x \cos \psi)^{\frac{1}{2}}} - \frac{10x(x - \cos \psi) + 7x^2}{(1 + x^2 - 2x \cos \psi)^{\frac{3}{2}}} \\
& + \frac{21x^2(x - \cos \psi)^2 + 9x^3(x - \cos \psi)}{(1 + x^2 - 2x \cos \psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos \psi)^3}{(1 + x^2 - 2x \cos \psi)^{\frac{7}{2}}}
\end{aligned}$$

Lengthy Derivations in Sect. 6.3.2

Basic Integrals

2.26 Forms containing $\sqrt{a + bx + cx^2}$ and integral powers of x

Notation: $R = a + bx + cx^2$, $\Delta = 4ac - b^2$

2.261¹¹ For $n = -1$

$$\int \frac{dx}{\sqrt{R}} = \frac{1}{\sqrt{c}} \ln \left(\frac{2\sqrt{cR} + 2cx + b}{\sqrt{\Delta}} \right) \quad [c > 0]$$

$$\int x\sqrt{R} dx = \frac{\sqrt{R^3}}{3c} - \frac{(2cx + b)b}{8c^2} \sqrt{R} - \frac{b\Delta}{16c^2} \int \frac{dx}{\sqrt{R}} \quad (\text{see 2.261})$$

2.264

$$1. \quad \int \frac{dx}{\sqrt{R}} \quad (\text{see 2.261})$$

$$2. \quad \int \frac{x dx}{\sqrt{R}} = \frac{\sqrt{R}}{c} - \frac{b}{2c} \int \frac{dx}{\sqrt{R}} \quad (\text{see 2.261})$$

$$3. \quad \int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x}{2c} - \frac{3b}{4c^2} \right) \sqrt{R} + \left(\frac{3b^2}{8c^2} - \frac{a}{2c} \right) \int \frac{dx}{\sqrt{R}} \quad (\text{see 2.261})$$

$$4. \quad \int \frac{x^3 dx}{\sqrt{R}} = \left(\frac{x^2}{3c} - \frac{5bx}{12c^2} + \frac{5b^2}{8c^3} - \frac{2a}{3c^2} \right) \sqrt{R} - \left(\frac{5b^3}{16c^3} - \frac{3ab}{4c^2} \right) \int \frac{dx}{\sqrt{R}} \quad (\text{see 2.261})$$

$$a = 1, \quad b = -2 \cos \Theta, \quad c = 1, \quad \Delta = 4(1 - \cos^2 \Theta) = 4 \sin^2 \Theta$$

$$\int x\sqrt{R} dx = \frac{1}{3}R^{\frac{3}{2}} - \frac{(2x - 2 \cos \Theta)(-2 \cos \Theta)}{8}R^{\frac{1}{2}} - \frac{(-2 \cos \Theta)(4 \sin^2 \Theta)}{16} \ln \left(\frac{2R^{\frac{1}{2}} + 2x - 2 \cos \Theta}{2 \sin \Theta} \right) + C$$

$$\int x\sqrt{R} dx = \frac{1}{3}R^{\frac{3}{2}} + \frac{(x - \cos \Theta) \cos \Theta}{2}R^{\frac{1}{2}} + \frac{\sin^2 \Theta \cos \Theta}{2} \ln \left(\frac{R^{\frac{1}{2}} + x - \cos \Theta}{\sin \theta} \right) + C$$

$\sin \theta$ in the denominator within the $\ln$ function contributes to a constant, and can be dropped.

$$\begin{aligned} \int x(1+x^2-2x\cos\theta)^{\frac{1}{2}} dx \\ = \frac{1}{3}(1+x^2-2x\cos\theta)^{\frac{3}{2}} + \frac{1}{2}(x-\cos\theta)\cos\theta(1+x^2-2x\cos\theta)^{\frac{1}{2}} \\ + \frac{1}{2}\sin^2\theta\cos\theta\ln\left[(1+x^2-2x\cos\theta)^{\frac{1}{2}}+x-\cos\theta\right] + C \end{aligned}$$

$$\int \frac{x^3 dx}{\sqrt{R}} = \left(\frac{x^2}{3} + \frac{5x\cos\theta}{6} + \frac{5\cos^2\theta}{2} - \frac{2}{3}\right)\sqrt{R} + \left(\frac{5\cos^3\theta}{2} - \frac{3\cos\theta}{2}\right)\ln\left(\frac{\sqrt{R}+x-\cos\theta}{\sin\theta}\right) + C$$

$$\begin{aligned} \int \frac{x^3 dx}{\sqrt{R}} &= \left(\frac{x^2}{3} + \frac{5x\cos\theta}{6} + \frac{5\cos^2\theta}{2} - \frac{2}{3}\right)(1+x^2-2x\cos\theta)^{\frac{1}{2}} \\ &+ \left(\frac{5\cos^3\theta}{2} - \frac{3\cos\theta}{2}\right)\ln\left[(1+x^2-2x\cos\theta)^{\frac{1}{2}}+x-\cos\theta\right] + C \end{aligned}$$

$$\int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x}{2} + \frac{3\cos\theta}{2}\right)\sqrt{R} + \left(\frac{3\cos^2\theta}{2} - \frac{1}{2}\right)\ln\left(\frac{\sqrt{R}+x-\cos\theta}{\sin\theta}\right)$$

$$\begin{aligned} \int \frac{x^2 dx}{\sqrt{R}} &= \left(\frac{x}{2} + \frac{3\cos\theta}{2}\right)(1+x^2-2x\cos\theta)^{\frac{1}{2}} \\ &+ \left(\frac{3\cos^2\theta}{2} - \frac{1}{2}\right)\ln\left[(1+x^2-2x\cos\theta)^{\frac{1}{2}}+x-\cos\theta\right] + C \end{aligned}$$

Eq. (6.117)

$$\begin{aligned} \int \frac{x^3 dx}{\sqrt{R}} - \cos\theta \int \frac{x^2 dx}{\sqrt{R}} \\ = \left(\frac{x^2}{3} + \frac{5x\cos\theta}{6} + \frac{5\cos^2\theta}{2} - \frac{2}{3} - \frac{x\cos\theta}{2} - \frac{3\cos^2\theta}{2}\right)\sqrt{R} \\ + \left(\frac{5\cos^3\theta}{2} - \frac{3\cos\theta}{2} - \frac{3\cos^3\theta}{2} + \frac{\cos\theta}{2}\right)\ln(\sqrt{R}+x-\cos\theta) + C \end{aligned}$$

$$\begin{aligned}
& \int \frac{x^3 dx}{\sqrt{R}} - \cos \Theta \int \frac{x^2 dx}{\sqrt{R}} \\
&= \left(\frac{x^2}{3} + \frac{x \cos \Theta}{3} + \cos^2 \Theta - \frac{2}{3} \right) \sqrt{x^2 - 2x \cos \Theta + 1} \\
&+ (\cos^3 \Theta - \cos \Theta) \ln \left(\sqrt{x^2 - 2x \cos \Theta + 1} + x - \cos \Theta \right) + C
\end{aligned}$$

$$\begin{aligned}
& \int \frac{x^3 dx}{\sqrt{R}} - \cos \Theta \int \frac{x^2 dx}{\sqrt{R}} \\
&= \left(\frac{x^2 + x \cos \Theta + 1}{3} - \sin^2 \Theta \right) \sqrt{x^2 - 2x \cos \Theta + 1} \\
&- \sin^2 \Theta \cos \Theta \ln \left(\sqrt{x^2 - 2x \cos \Theta + 1} + x - \cos \Theta \right) + C
\end{aligned}$$

$$\begin{aligned}
& \int \frac{x^3 dx}{\sqrt{R}} - \cos \Theta \int \frac{x^2 dx}{\sqrt{R}} \\
&= \left(\frac{x^2 + x \cos \Theta + 1}{3} - \sin^2 \Theta \right) \sqrt{x^2 - 2x \cos \Theta + 1} \\
&- \sin^2 \Theta \cos \Theta \ln \left(\sqrt{x^2 - 2x \cos \Theta + 1} + x - \cos \Theta \right) + C
\end{aligned}$$

$$\begin{aligned}
& \int \frac{x^3 dx}{\sqrt{R}} - \cos \Theta \int \frac{x^2 dx}{\sqrt{R}} + \int x^2 dx \\
&= \frac{1}{3} x^3 + \left(\frac{x^2 + x \cos \Theta + 1}{3} - \sin^2 \Theta \right) \sqrt{x^2 - 2x \cos \Theta + 1} \\
&- \sin^2 \Theta \cos \Theta \ln \left(\sqrt{x^2 - 2x \cos \Theta + 1} + x - \cos \Theta \right) + C
\end{aligned}$$

$$\begin{aligned}
& \int \frac{x^3 dx}{\sqrt{R}} - \cos \Theta \int \frac{x^2 dx}{\sqrt{R}} + \int x^2 dx \\
&= \frac{1}{3} \left[x^3 + (x^2 + x \cos \Theta + 1 - 3 \sin^2 \Theta) \sqrt{x^2 - 2x \cos \Theta + 1} \right. \\
&\left. - 3 \sin^2 \Theta \cos \Theta \ln \left(\sqrt{x^2 - 2x \cos \Theta + 1} + x - \cos \Theta \right) \right] + C
\end{aligned}$$

Eq. (6.124)—(6.126)

$$\begin{aligned}
f(x) &= \frac{1}{3} (1 + x^2 - 2x \cos \Theta)^{\frac{3}{2}} + \frac{1}{2} (x - \cos \Theta) \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} - \frac{1}{2} x^2 + \frac{1}{3} x^3 \\
&+ \frac{1}{2} \sin^2 \Theta \cos \Theta \ln \left[(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} + x - \cos \Theta \right]
\end{aligned}$$

$$\begin{aligned}
f'(x) &= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} + \frac{(x - \cos \Theta)^2 \cos \Theta}{2(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} \\
&\quad - x + x^2 + \frac{1}{2} \sin^2 \Theta \cos \Theta \frac{\frac{(x - \cos \Theta)}{(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} + 1}{(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} + x - \cos \Theta} \\
&= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} \\
&\quad + \frac{(x - \cos \Theta)^2 \cos \Theta}{2(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - x + x^2 + \frac{\sin^2 \Theta \cos \Theta}{2(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} \\
&= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} \\
&\quad + \frac{[(x - \cos \Theta)^2 + \sin^2 \Theta] \cos \Theta}{2(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - x + x^2 \\
&= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} \\
&\quad + \frac{[x^2 - 2x \cos \Theta + \cos^2 \Theta + \sin^2 \Theta] \cos \Theta}{2(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - x + x^2 \\
&= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} \\
&\quad + \frac{[x^2 - 2x \cos \Theta + 1] \cos \Theta}{2(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - x + x^2 \\
&= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} \\
&\quad + \frac{1}{2} \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} - x + x^2 \\
&= (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}(x - \cos \Theta) + \cos \Theta (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} - x + x^2 \\
&= x(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} - x + x^2
\end{aligned}$$

$$f''(x) = (1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}} + \frac{x(x - \cos \Theta)}{(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - 1 + 2x$$

$$\begin{aligned}
f'''(x) &= \frac{x - \cos \Theta}{(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} + \frac{(x - \cos \Theta) + x}{(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - \frac{x(x - \cos \Theta)^2}{(1 + x^2 - 2x \cos \Theta)^{\frac{3}{2}}} + 2 \\
&= \frac{x + 2(x - \cos \Theta)}{(1 + x^2 - 2x \cos \Theta)^{\frac{1}{2}}} - \frac{x(x - \cos \Theta)^2}{(1 + x^2 - 2x \cos \Theta)^{\frac{3}{2}}} + 2
\end{aligned}$$

Eq. (6.127)—(6.129)

$$\begin{aligned}
 3f(x) &= \int \frac{x^3 dx}{\sqrt{R}} - \cos \theta \int \frac{x^2 dx}{\sqrt{R}} + \int x^2 dx \\
 &= (x^2 + x \cos \theta + 1 - 3 \sin^2 \theta) \sqrt{x^2 - 2x \cos \theta + 1} + x^3 \\
 &\quad - 3 \sin^2 \theta \cos \theta \ln \left(\sqrt{x^2 - 2x \cos \theta + 1} + x - \cos \theta \right)
 \end{aligned}$$

$$\begin{aligned}
 3f'(x) &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + \frac{(x^2 + x \cos \theta + 1 - 3 \sin^2 \theta)(x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + 3x^2 \\
 &\quad - 3 \sin^2 \theta \cos \theta \frac{\frac{x - \cos \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} + 1}{\sqrt{x^2 - 2x \cos \theta + 1} + x - \cos \theta} \\
 &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + \frac{(x^2 + x \cos \theta + 1 - 3 \sin^2 \theta)(x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + 3x^2 \\
 &\quad - \frac{3 \sin^2 \theta \cos \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 \\
 &\quad + \frac{(x^2 + x \cos \theta + 1)(x - \cos \theta) - 3 \sin^2 \theta (x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} - \frac{3 \sin^2 \theta \cos \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 \\
 &\quad + \frac{(x^2 + x \cos \theta + 1)(x - \cos \theta) - 3x \sin^2 \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 \\
 &\quad + \frac{(x^2 - 2x \cos \theta + 1)(x - \cos \theta) + 3x \cos \theta (x - \cos \theta) - 3x \sin^2 \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 + (x - \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} \\
 &\quad + \frac{3x \cos \theta (x - \cos \theta) - 3x \sin^2 \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= (2x + \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + (x - \cos \theta) \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 \\
 &\quad + \frac{3x^2 \cos \theta - 3x \cos^2 \theta - 3x \sin^2 \theta}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= 3x \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 + \frac{3x^2 \cos \theta - 3x}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
 &= 3x \sqrt{x^2 - 2x \cos \theta + 1} + 3x^2 + \frac{3x(x \cos \theta - 1)}{\sqrt{x^2 - 2x \cos \theta + 1}} \text{ (used for } f'') \\
 &= 3x^2 + 3x \sqrt{x^2 - 2x \cos \theta + 1} - \frac{3x(1 - x \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}}
 \end{aligned}$$

$$\begin{aligned}
3f''(x) &= 3\sqrt{x^2 - 2x \cos \theta + 1} + \frac{3x(x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + 6x + \frac{6x \cos \theta - 3}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
&\quad - \frac{3x(x \cos \theta - 1)(x - \cos \theta)}{(\sqrt{x^2 - 2x \cos \theta + 1})^3} \\
&= 3\sqrt{x^2 - 2x \cos \theta + 1} + \frac{3x(x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + 6x + \frac{6x \cos \theta - 3 - 3x^2}{\sqrt{x^2 - 2x \cos \theta + 1}} \\
&\quad + \frac{3x^2}{\sqrt{x^2 - 2x \cos \theta + 1}} - \frac{3x(x \cos \theta - 1)(x - \cos \theta)}{(\sqrt{x^2 - 2x \cos \theta + 1})^3} \\
&= 3\sqrt{x^2 - 2x \cos \theta + 1} + \frac{3x(x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + 6x - 3\sqrt{x^2 - 2x \cos \theta + 1} \\
&\quad + \frac{3x^2}{\sqrt{x^2 - 2x \cos \theta + 1}} - \frac{3x(x \cos \theta - 1)(x - \cos \theta)}{(\sqrt{x^2 - 2x \cos \theta + 1})^3} \\
&= 6x + \frac{3x(x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + \frac{3x^2}{\sqrt{x^2 - 2x \cos \theta + 1}} - \frac{3x(x \cos \theta - 1)(x - \cos \theta)}{(\sqrt{x^2 - 2x \cos \theta + 1})^3} \\
&= 6x + \frac{3x(2x - \cos \theta)}{\sqrt{x^2 - 2x \cos \theta + 1}} + \frac{3x(1 - x \cos \theta)(x - \cos \theta)}{(\sqrt{x^2 - 2x \cos \theta + 1})^3}
\end{aligned}$$

$$\begin{aligned}
3f'''(x) &= 6 + \frac{3(2x - \cos \theta) + 6x}{(x^2 - 2x \cos \theta + 1)^{1/2}} - \frac{3x(2x - \cos \theta)(x - \cos \theta)}{(x^2 - 2x \cos \theta + 1)^{\frac{3}{2}}} \\
&\quad + \frac{3(1 - x \cos \theta)(x - \cos \theta) - 3x \cos \theta (x - \cos \theta) + 3x(1 - x \cos \theta)}{(x^2 - 2x \cos \theta + 1)^{\frac{3}{2}}} - \frac{9x(1 - x \cos \theta)(x - \cos \theta)^2}{(x^2 - 2x \cos \theta + 1)^{\frac{5}{2}}} \\
&= 6 + \frac{3(x - \cos \theta) + 9x}{(x^2 - 2x \cos \theta + 1)^{1/2}} \\
&\quad + \frac{3(1 - x \cos \theta)(x - \cos \theta) - 3x \cos \theta (x - \cos \theta) + 3x(1 - x \cos \theta) - 3x(2x - \cos \theta)(x - \cos \theta)}{(x^2 - 2x \cos \theta + 1)^{\frac{3}{2}}} \\
&\quad - \frac{9x(1 - x \cos \theta)(x - \cos \theta)^2}{(x^2 - 2x \cos \theta + 1)^{\frac{5}{2}}} \\
&= 6 + \frac{3(x - \cos \theta) + 9x}{(x^2 - 2x \cos \theta + 1)^{1/2}} \\
&\quad + \frac{3(1 - x \cos \theta)(x - \cos \theta) + 3x(1 - x \cos \theta) - 3x(2x - \cos \theta)(x - \cos \theta) - 3x \cos \theta (x - \cos \theta)}{(x^2 - 2x \cos \theta + 1)^{\frac{3}{2}}} \\
&\quad - \frac{9x(1 - x \cos \theta)(x - \cos \theta)^2}{(x^2 - 2x \cos \theta + 1)^{\frac{5}{2}}}
\end{aligned}$$

$$\begin{aligned}
&= 6 + \frac{3(x - \cos \theta) + 9x}{(x^2 - 2x \cos \theta + 1)^{1/2}} \\
&+ \frac{3(1 - x \cos \theta)(x - \cos \theta) + 3x(1 - x \cos \theta) - 3x(x - \cos \theta)[(2x - \cos \theta) + \cos \theta]}{(x^2 - 2x \cos \theta + 1)^{\frac{3}{2}}} \\
&- \frac{9x(1 - x \cos \theta)(x - \cos \theta)^2}{(x^2 - 2x \cos \theta + 1)^{\frac{5}{2}}} \\
&= 6 + \frac{3(x - \cos \theta) + 9x}{(x^2 - 2x \cos \theta + 1)^{1/2}} + \frac{3(1 - x \cos \theta)(x - \cos \theta) + 3x(1 - x \cos \theta) - 6x^2(x - \cos \theta)}{(x^2 - 2x \cos \theta + 1)^{\frac{3}{2}}} \\
&- \frac{9x(1 - x \cos \theta)(x - \cos \theta)^2}{(x^2 - 2x \cos \theta + 1)^{\frac{5}{2}}}
\end{aligned}$$

Eq. (6.131)–(6.133)

$$f'(1) = x(1 + x^2 - 2x \cos \theta)^{\frac{1}{2}} - x + x^2 = (2 - 2 \cos \theta)^{\frac{1}{2}} = 2 \sin \frac{\theta}{2}$$

$$\begin{aligned}
f''(1) &= (1 + x^2 - 2x \cos \theta)^{\frac{1}{2}} + \frac{x(x - \cos \theta)}{(1 + x^2 - 2x \cos \theta)^{\frac{1}{2}}} - 1 + 2x \\
&= (2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{(1 - \cos \theta)}{(2 - 2 \cos \theta)^{\frac{1}{2}}} - 1 + 2 = (2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{(2 - 2 \cos \theta)}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} + 1 \\
&= 1 + (2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{1}{2}(2 - 2 \cos \theta)^{\frac{1}{2}} = 1 + \frac{3}{2}(2 - 2 \cos \theta)^{\frac{1}{2}} = 1 + 3 \sin \frac{\theta}{2}
\end{aligned}$$

$$\begin{aligned}
f'''(1) &= \frac{x}{(1 + x^2 - 2x \cos \theta)^{\frac{1}{2}}} + \frac{2(x - \cos \theta)}{(1 + x^2 - 2x \cos \theta)^{\frac{1}{2}}} - \frac{x(x - \cos \theta)^2}{(1 + x^2 - 2x \cos \theta)^{\frac{3}{2}}} + 2 \\
&= \frac{1}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{(2 - 2 \cos \theta)}{(2 - 2 \cos \theta)^{\frac{1}{2}}} - \frac{(1 - \cos \theta)^2}{(2 - 2 \cos \theta)^{\frac{3}{2}}} + 2 \\
&= \frac{1}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + (2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{(2 - 2 \cos \theta)^2}{4(2 - 2 \cos \theta)^{\frac{3}{2}}} + 2 \\
&= \frac{1}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + (2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{1}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} + 2 \\
&= 2 + \frac{1}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} = 2 + \frac{1}{2 \sin \frac{\theta}{2}} + \frac{3}{2} \sin \frac{\theta}{2} \\
&= \frac{1}{2 \sin \frac{\theta}{2}} + 2 + \frac{3}{2} \sin \frac{\theta}{2} = \frac{1}{2 \sin \frac{\theta}{2}} \left(1 + 4 \sin \frac{\theta}{2} + 3 \sin^2 \frac{\theta}{2} \right)
\end{aligned}$$

Eq. (6.134)—(6.136)

$$\begin{aligned}
 3f'(x) &= 3x^2 + 3x\sqrt{x^2 - 2x\cos\theta + 1} - \frac{3x(1 - x\cos\theta)}{\sqrt{x^2 - 2x\cos\theta + 1}} = 3 + 3(2 - \cos\theta)^{\frac{1}{2}} - \frac{3(1 - \cos\theta)}{(2 - \cos\theta)^{\frac{1}{2}}} = \\
 &= 3 + 3(2 - \cos\theta)^{\frac{1}{2}} - \frac{3(2 - 2\cos\theta)}{2(2 - \cos\theta)^{\frac{1}{2}}} = 3 + 3(2 - \cos\theta)^{\frac{1}{2}} - \frac{3}{2}(2 - \cos\theta)^{\frac{1}{2}} \\
 &= 3 + \frac{3}{2}(2 - \cos\theta)^{\frac{1}{2}} = 3 + 3\sin\frac{\theta}{2}
 \end{aligned}$$

$$\begin{aligned}
 3f''(1) &= 6x + \frac{3x(2x - \cos\theta)}{\sqrt{x^2 - 2x\cos\theta + 1}} + \frac{3x(1 - x\cos\theta)(x - \cos\theta)}{(\sqrt{x^2 - 2x\cos\theta + 1})^3} \\
 &= 6 + \frac{3(2 - \cos\theta)}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(1 - \cos\theta)^2}{(2 - \cos\theta)^{\frac{3}{2}}} \\
 &= 6 + \frac{3}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(1 - \cos\theta)}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(1 - \cos\theta)^2}{(2 - \cos\theta)^{\frac{3}{2}}} \\
 &= 6 + \frac{3}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\theta)}{2(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\theta)^2}{4(2 - \cos\theta)^{\frac{3}{2}}} \\
 &= 6 + \frac{3}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\theta)}{2(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\theta)^2}{4(2 - \cos\theta)^{\frac{3}{2}}} \\
 &= 6 + \frac{3}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{3}{2}(2 - \cos\theta)^{\frac{1}{2}} + \frac{3}{4}(2 - \cos\theta)^{\frac{1}{2}} \\
 &= 6 + \frac{3}{(2 - \cos\theta)^{\frac{1}{2}}} + \frac{9}{4}(2 - \cos\theta)^{\frac{1}{2}} = 6 + \frac{3}{2\sin\frac{\theta}{2}} + \frac{9}{2}\sin\frac{\theta}{2}
 \end{aligned}$$

$$\begin{aligned}
 3f'''(1) &= 6 + \frac{3(x - \cos\theta) + 9x}{(x^2 - 2x\cos\theta + 1)^{1/2}} \\
 &\quad + \frac{3(1 - x\cos\theta)(x - \cos\theta) + 3x(1 - x\cos\theta) - 6x^2(x - \cos\theta)}{(x^2 - 2x\cos\theta + 1)^{\frac{3}{2}}} \\
 &\quad - \frac{9x(1 - x\cos\theta)(x - \cos\theta)^2}{(x^2 - 2x\cos\theta + 1)^{\frac{5}{2}}} \\
 &= 6 + \frac{3(1 - \cos\theta) + 9}{(2 - 2\cos\theta)^{1/2}} + \frac{3(1 - \cos\theta)^2 + 3(1 - \cos\theta) - 6(1 - \cos\theta)}{(2 - 2\cos\theta)^{\frac{3}{2}}} \\
 &\quad - \frac{9(1 - \cos\theta)^3}{(2 - 2\cos\theta)^{\frac{5}{2}}}
 \end{aligned}$$

$$\begin{aligned}
&= 6 + \frac{3(1 - \cos \theta) + 9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3(1 - \cos \theta)^2 - 3(1 - \cos \theta)}{(2 - 2 \cos \theta)^{\frac{3}{2}}} - \frac{9(x - \cos \theta)^3}{(2 - 2 \cos \theta)^{\frac{5}{2}}} \\
&= 6 + \frac{3(2 - 2 \cos \theta)}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3(2 - 2 \cos \theta)^2}{4(2 - 2 \cos \theta)^{\frac{3}{2}}} - \frac{3(2 - 2 \cos \theta)}{2(2 - 2 \cos \theta)^{\frac{3}{2}}} \\
&\quad - \frac{9(2 - 2 \cos \theta)^3}{8(2 - 2 \cos \theta)^{\frac{5}{2}}} \\
&= 6 + \frac{3}{2}(2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{3}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} \\
&\quad - \frac{9}{8}(2 - 2 \cos \theta)^{\frac{1}{2}} \\
&= 6 + \frac{3}{2}(2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{3}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{9}{8}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{3}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} \\
&\quad + \frac{9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} \\
&= 6 + \frac{9}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{9}{8}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{3}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{18}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} \\
&= 6 + \frac{9}{8}(2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{15}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} = 6 + \frac{9}{4} \sin \frac{\theta}{2} + \frac{15}{4 \sin \frac{\theta}{2}} = 6 + \frac{9a_r}{8r} + \frac{15r}{2a_r} \\
&= \frac{15r}{2a_r} \left(1 + \frac{4a_r}{5r} + \frac{3a_r^2}{20r^2} \right) \\
&= 6 + \frac{3(1 - \cos \theta) + 9}{(2 - 2 \cos \theta)^{1/2}} + \frac{3(1 - \cos \theta)^2 + 3(1 - \cos \theta) - 6(1 - \cos \theta)}{(2 - 2 \cos \theta)^{\frac{3}{2}}} \\
&\quad - \frac{9(1 - \cos \theta)^3}{(2 - 2 \cos \theta)^{\frac{5}{2}}} \\
&= 6 + \frac{3(1 - \cos \theta) + 9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3(1 - \cos \theta)^2 - 3(1 - \cos \theta)}{(2 - 2 \cos \theta)^{\frac{3}{2}}} - \frac{9(x - \cos \theta)^3}{(2 - 2 \cos \theta)^{\frac{5}{2}}} \\
&= 6 + \frac{3(2 - 2 \cos \theta)}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3(2 - 2 \cos \theta)^2}{4(2 - 2 \cos \theta)^{\frac{3}{2}}} - \frac{3(2 - 2 \cos \theta)}{2(2 - 2 \cos \theta)^{\frac{3}{2}}} \\
&\quad - \frac{9(2 - 2 \cos \theta)^3}{8(2 - 2 \cos \theta)^{\frac{5}{2}}} \\
&= 6 + \frac{3}{2}(2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{9}{(2 - 2 \cos \theta)^{\frac{1}{2}}} + \frac{3}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{3}{2(2 - 2 \cos \theta)^{\frac{1}{2}}} \\
&\quad - \frac{9}{8}(2 - 2 \cos \theta)^{\frac{1}{2}} \\
&= 6 + \frac{3}{2}(2 - 2 \cos \theta)^{\frac{1}{2}} + \frac{3}{4}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{9}{8}(2 - 2 \cos \theta)^{\frac{1}{2}} - \frac{3}{2(2 - 2 \cos \theta)^{\frac{1}{2}}}
\end{aligned}$$

$$\begin{aligned}
& + \frac{9}{(2 - 2 \cos \Theta)^{\frac{1}{2}}} \\
& = 6 + \frac{9}{4}(2 - 2 \cos \Theta)^{\frac{1}{2}} - \frac{9}{8}(2 - 2 \cos \Theta)^{\frac{1}{2}} - \frac{3}{2(2 - 2 \cos \Theta)^{\frac{1}{2}}} + \frac{18}{2(2 - 2 \cos \Theta)^{\frac{1}{2}}} \\
& = 6 + \frac{9}{8}(2 - 2 \cos \Theta)^{\frac{1}{2}} + \frac{15}{2(2 - 2 \cos \Theta)^{\frac{1}{2}}} = 6 + \frac{9}{4} \sin \frac{\Theta}{2} + \frac{15}{4 \sin \frac{\Theta}{2}} = 6 + \frac{9a_r}{8r} + \frac{15r}{2a_r} \\
& = \frac{15r}{2a_r} \left(1 + \frac{4a_r}{5r} + \frac{3a_r^2}{20r^2} \right)
\end{aligned}$$